

El Houcine El Yakoubi

DevSecOps & Infrastructure Security Architect

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Technical Palette

Container Orchestration & CI/CD

Docker, Kubernetes (K3s/K3d), Helm, ArgoCD, GitOps, Jenkins, GitLab CI, Terraform, Ansible, Nix, Bazel

Security Engineering

SAST/DAST (SonarQube, Trivy, OWASP ZAP), Pentesting, LLM Security, ASVS, STRIDE Threat Modeling, Incident Response, CVE Analysis

Infra Security & Compliance

HashiCorp Vault, OpenSCAP (PCI-DSS, ISO 27017), ModSecurity, Suricata, Snort, ELK SIEM, Falco, OPA, SELinux/AppArmor, PKI/TLS, HSM, Secure Boot

Cloud & Monitoring

AWS, Proxmox, Nginx, Prometheus, Grafana, Datadog

Languages & Data

C/C++, Zig, Python, Bash, Git, SQL (PostgreSQL), Supabase, ORMs

Embedded & Industrial

PLC, SCADA/HMI, KiCad, CAN, UART/I2C/SPI, BLE/Zigbee/LoRa/NFC

Education

Network & Information Systems Architect

1337 School (42 Network), UM6P Ben Guerir
Expected 2027 | [RNCP Level 7](#) (Master's Equivalent)

Certifications

Google Cybersecurity Certificate: Foundational -- Google Career Certificates

Google Cloud Professional Cloud DevOps Engineer: CI/CD, SRE, monitoring on GCP

Offensive Security: 20+ HTB/THM challenges -- hardware, crypto, RE, OSINT

Cloud Security: Global 42 AWS Cloud Quest Tournament

Languages

English (Fluent), Arabic (Native)

Leadership & Affiliations

President | Leet-Makers Robotics Club Nov 2022 – Present

- ▶ Led robotics security workshops for 150+ learners blending embedded hardening with secure firmware delivery.

Technical Member & CTF Developer | Elitesec Club May 2023 – Present

- ▶ Architected OSINT/hardware/crypto CTF tracks and automated SecDevOps workflows via GitOps and policy-as-code.
- ▶ Led OPSEC and hardware-hacking masterclasses; codified security-as-code templates to scale collaboration.

Professional Summary

DevSecOps professional with hands-on corporate experience in cloud-native security, autonomous infrastructure (HashiCorp Vault, Kubernetes), and AI-driven pentesting. Proven track record of reducing secrets sprawl across a sovereign cloud provider's K3s infrastructure via Vault Agent sidecar injection and dynamic PostgreSQL credential rotation, while enforcing compliance (PCI-DSS, ISO 27017) across production environments.

Corporate Experience

DevSecOps Intern — Cloud/Infra Team | Atlas Cloud Services (ACS) 6 Months • On-site

- ▶ Delivered VMaaS “Zero-Touch” automation across DNS, backup, monitoring, and tagging workflows for a sovereign cloud provider.
- ▶ Reduced secrets sprawl across K3s infrastructure via HashiCorp Vault Agent sidecar injection and dynamic PostgreSQL credential rotation.
- ▶ Managed PKI/TLS lifecycle and automated certificate rotation workflows.
- ▶ Conducted compliance auditing using OpenSCAP to align infrastructure with PCI-DSS and ISO 27017 frameworks.
- ▶ Integrated Vault Radar to proactively identify and remediate hardcoded secrets across repositories.

DevSecOps Intern — AI Security Automation | GSNA Solutions 6 Months • Hybrid

- ▶ Architected from the ground up an LLM-powered automated pentesting orchestrator in Python, integrating Nmap, Nikto, and Metasploit for autonomous reconnaissance and exploitation -- reducing manual pentesting effort by ~73%.
- ▶ Implemented end-to-end vulnerability analysis: IP-in/report-out pipeline synchronizing scan data with the NVD API (CVE scoring) and leveraging AI for structured, actionable reporting.
- ▶ Embedded SAST/DAST gates (SonarQube, Trivy, OWASP ZAP) into CI/CD pipelines and containerized the testing suite using Docker.

Key Projects

- ▶ **Web App & DevSecOps** | **ft_transcendence**: Multiplayer Django app secured with ModSecurity WAF, ELK SIEM, Vault, Prometheus/Grafana, and automated CI vulnerability scanning.
- ▶ **Secure LEMP Stack** | **Inception**: Nginx SSL/TLS 1.3 reverse proxy, encrypted MariaDB, PHP-FPM, Redis caching, and isolated FTP in Docker.
- ▶ **Linux Hardening** | **Born2beRoot**: Certificate-based SSH, UFW, AppArmor, LVM encryption, and cron monitoring on hardened Debian.
- ▶ **Kubernetes IoT Lab** | **Inception-of-Things**: K3s/K3d lab with Ingress, Argo CD, and GitLab-driven GitOps automation.
- ▶ **Ansible Orchestrator** | **cloud-1**: Tag-aware Ansible roles and vault workflows for repeatable remote cloud deployments.
- ▶ **Homelab & Declarative Infra**: Proxmox/TrueNAS private cloud with declarative NixOS services and an OpenBSD packet-filter firewall.
- ▶ **Robotics & Embedded Security**: Hardened stacks from KiCad PCBs to C/C++ firmware with resilient I2C/SPI buses and secure boot paths.

Research Interests

Zero-knowledge & trust proofs, post-quantum crypto implementation, embedded supply chain security, OSINT automation, hardware side-channel analysis, and open-source security tooling.